XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

2026/2027 MAT STUDENT COUNCIL

EVENT FLOW PLAN

EVENT NAME	2609 Semester MAT × Library Mathematics Week
EVENT DATE	Week 4 of 2609 Semester (19th October 2026 to 25th October 2026)
EVENT TIME	Library Opening Hours
EVENT VENUE	Library 1st,2nd,3rd floor
ORGANISER	Ooi Ray Shua

Strategy and Goals:

- **Foot-Traffic Strategy**: Use interactive mini-booths as engaging "teasers" to attract crowds, naturally guiding them into the main Exhibition which serves as the complete, in-depth showcase of the event's mathematics.
- **Core Message:** Showcase **(Power of Math)** as visually appealing and practically relevant across all university majors—not just boring paperwork.
- **Reusability:** Build durable, evergreen props and displays that the student council can easily reuse for future orientations and events.

Scope Reminder: This is Math Week, not MAT Department Week. Keep the content centered on celebrating mathematics broadly rather than over-promoting our specific program.

Role	Name	Job Scope
Advisor	Dr. Peter Zeiner	Offer advice for organizing events.
Chairperson	Ooi Ray Shua	In charge of personnel arrangement and work division.
Vice Chairperson	Indayang Fyna Desyna	Assist Chairperson.
	Chew Yee Xuan	
	Lim Yen Boei	
Secretary	Oh Bing Xiong	Record meeting minutes and program (chain of events) summary.
	Pang Kenwy	
	Xie Wanyan	
Treasurer	Liew Huei Qi	Manage the finance when necessary.

Event Planner Team		
Role	Name	Job Scope
Coordinator	Chew Yee Xuan	Oversee and assist chairpersons in carrying out their roles.
	Lim Yen Boei	
Exhibition Team	Chu Yew Xuan	Curate theme-aligned mathematical concepts and assemble foundational imagery and layouts for a gallery-style exhibition, collaborating with the design team for final artistic polish and refinement
	Cher Ming Sian	
MAT Mini Booth Team	Chen Chen	Develop the digital platform and design interactive, educational mini-games and puzzles that balance high engagement with mathematical learning.
	Kow Chun Xiang	
	Chan Yu Hwong	
	Lim Xin Yong	
MAT Mini Event Team	Lew Wai Hong	Lead the organization of a featured Math Week event, acting as the primary point of contact and liaison for all inter-club collaborations if any.
	Jason William Teja	
	Lee Shuan Fung	

Refer to 2610_MathWeek_EventPlanerTask.xlsx for more details.

Exhibition/Props Production Team		
Role	Name	Job Scope
Coordinator	Indayang Fyna Desyna	Oversee and assist chairpersons in carrying out their roles.
General Affairs	Choo Boon Ping	Prepare the props needed for the event.
	Soh Shiuan Ming	
	Cliff Pramana	
Design	Chua Chuek Eu	Design promotional materials for the event.
	Kang Hong Jie	
	Siow Shi Hui	

Multimedia Team		
Role	Name	Job Scope
Coordinator	Indayang Fyna Desyna	Oversee and assist chairpersons in carrying out their roles.
Design	Chua Chuek Eu	Design promotional materials for the event.
	Kang Hong Jie	
	Siow Shi Hui	
Coordinator	Lim Yen Boei	Oversee and assist chairpersons in carrying out their roles.
Public Relations	Eunice Khoo Kai Xin	Manage and promote the event on the Student Council's social media.
	Gan Ruoxian	
	Choo Jin Sheng	
Coordinator	Chew Yee Xuan	Oversee and assist chairpersons in carrying out their roles.
Photography & Videography	Wendy Kua Rou Yin	Shooting & recording during the event, and editing of the promotional video.
	Chen Jingyi	
	Lim Wey Jun	

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Event General Details

Math Week Themes

Each block of days is themed and aligned to a group of academic programmes:

Days / Theme	Mathematics in **Associated Programmes**
Mon – Wed Mathematics in Finance / Game / Logic Puzzles	• Economics in Finance • Management in Accounting • Management in International Business • Management in E-Commerce • Management in Hospitality Management
Tue – Thu Mathematics in AI and Data Science	• Artificial Intelligence • Data Science • Computer Science and Technology • Software Engineering • Digital Media Technology • Cyber Security
Wed – Fri Mathematics in Arts / Literature / TCM Medical Science	• Traditional Chinese Medicine • English Language and Literature • Chinese Studies • Communication • Journalism • Advertising
Fri – Sun Mathematics in Engineering / Sciences / Math	• Mathematics and Applied Mathematics • Physics • Electrical and Electronics Engineering • Robotics and Automation Engineering • Chemical Engineering • New Energy Science and Engineering • Marine Biotechnology • Marine Environmental Chemistry

Need to include foundation?

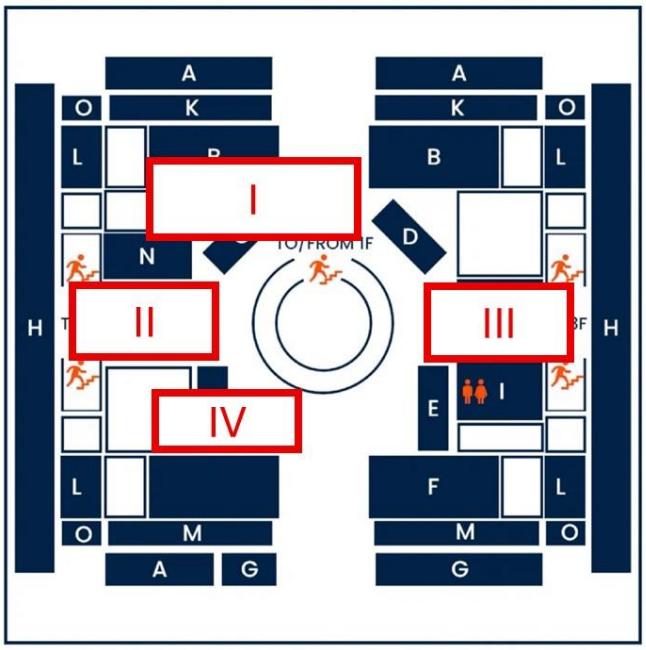
Itinerary of Math Week

Pre - Math Week (Sunday)

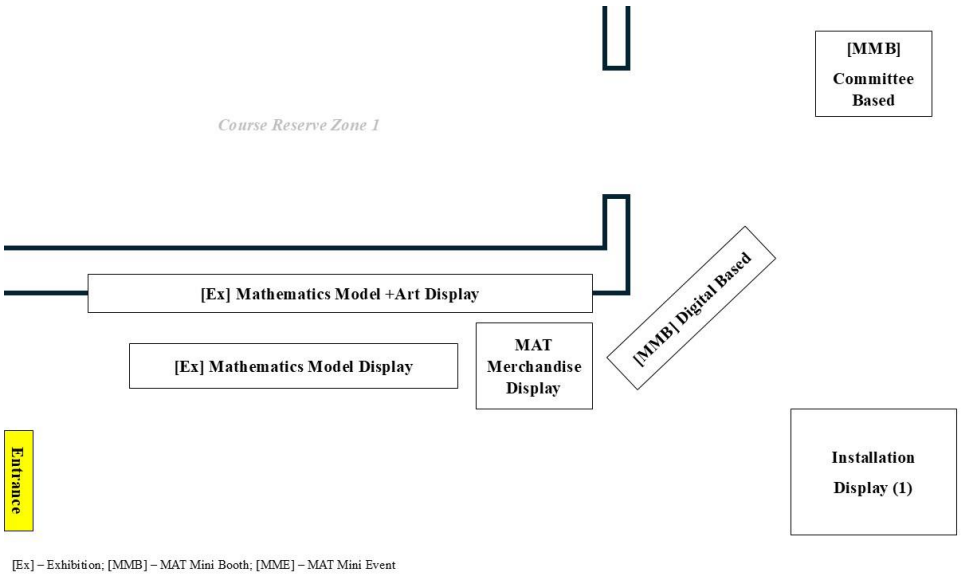
Time		Activity	PIC
xx:xx	xx:xx	{}	{}
Whole Event		Photo and video taking	{}

Floor Plan

Library 2nd Floor

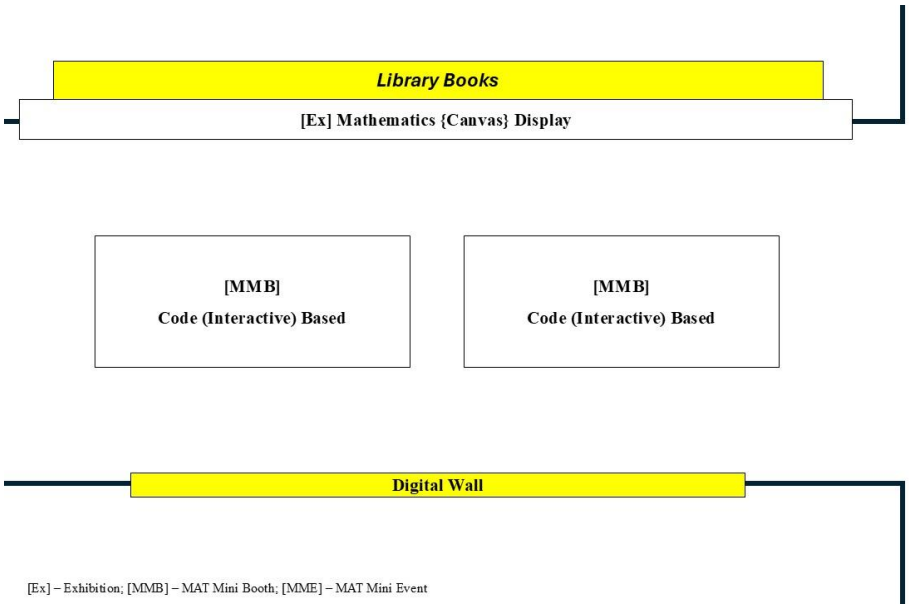


(I)

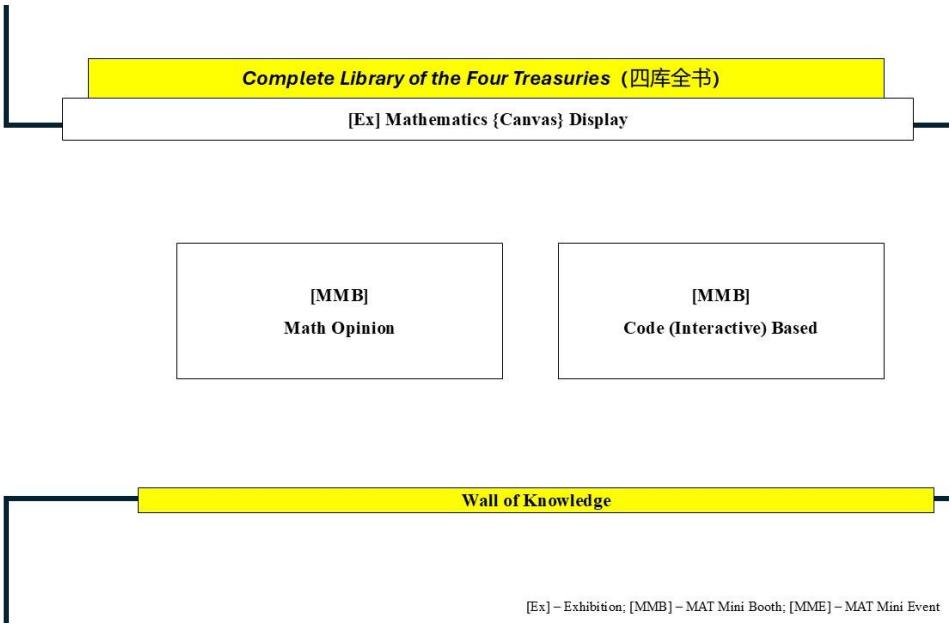


[Ex] – Exhibition; [MMB] – MAT Mini Booth; [MME] – MAT Mini Event

(II)

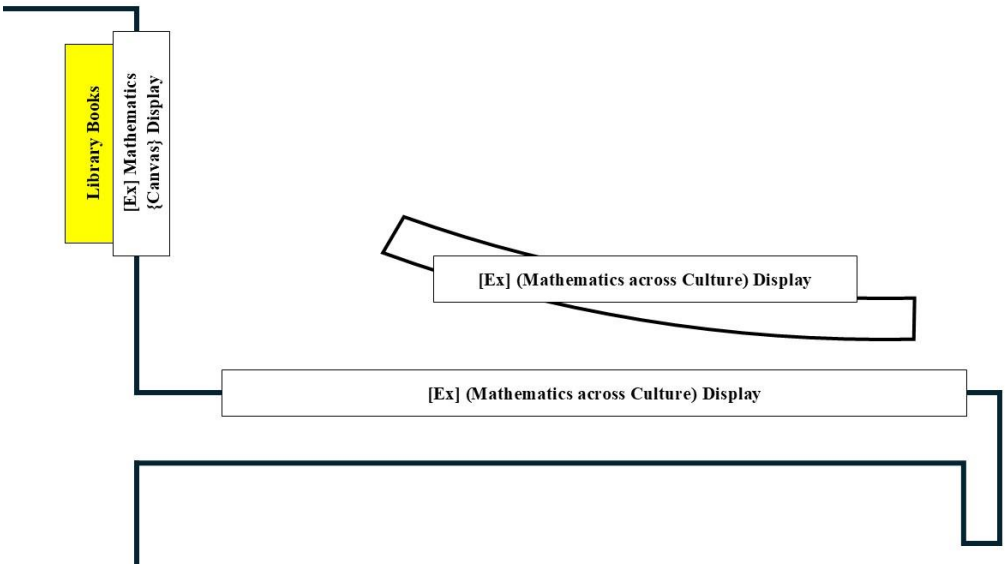

Library Books
[Ex] Mathematics {Canvas} Display
[MMB]
Code (Interactive) Based
[MMB]
Code (Interactive) Based
Digital Wall
[Ex] – Exhibition; [MMB] – MAT Mini Booth; [MME] – MAT Mini Event

(III)

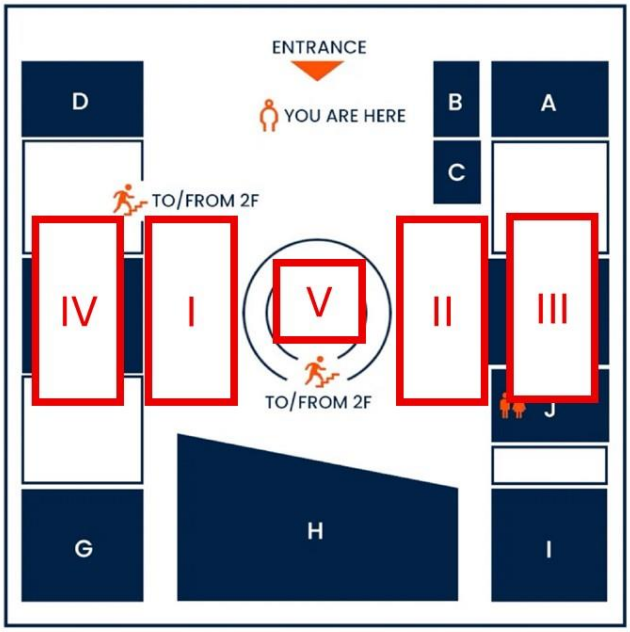

Complete Library of the Four Treasuries（四库全书）
[Ex] Mathematics {Canvas} Display
[MMB]
Math Opinion
[MMB]
Code (Interactive) Based
Wall of Knowledge
[Ex] – Exhibition; [MMB] – MAT Mini Booth; [MME] – MAT Mini Event

(IV)

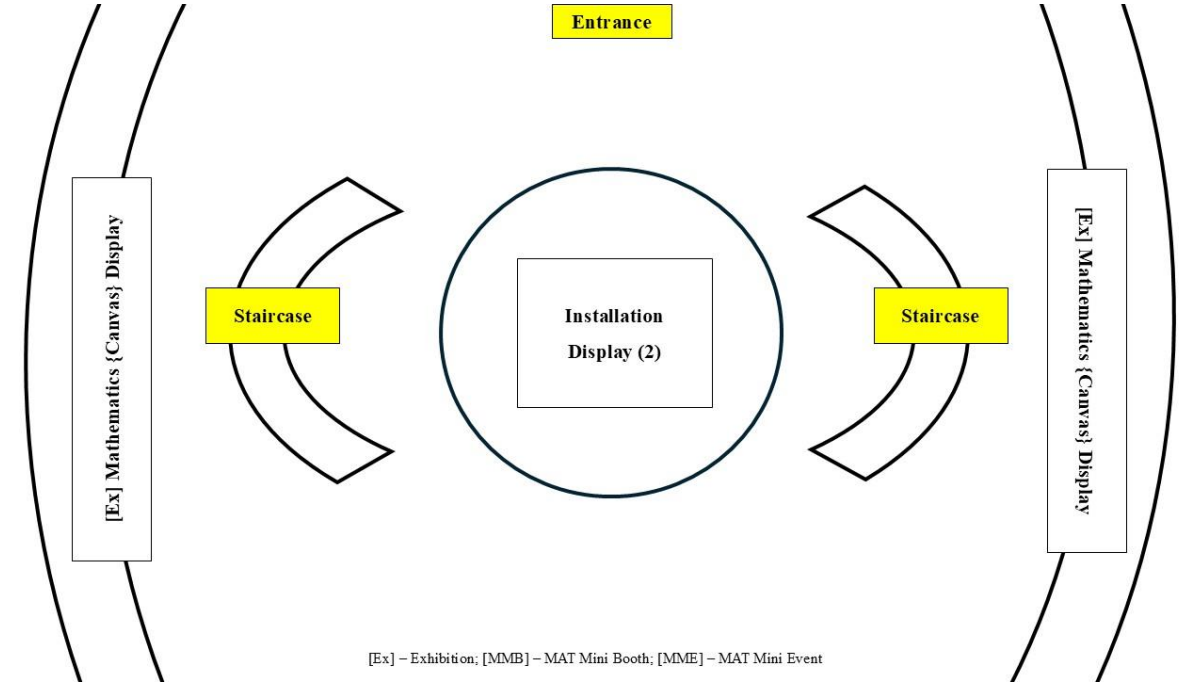
[Ex] – Exhibition; [MMB] – MAT Mini Booth; [MME] – MAT Mini Event



Library 1st Floor



(I,V,II)



Event Activity Details

Merchandise Display

The MAT Student Council's own merchandise is displayed at the merchandise counter beside the Digital / Committee booths (Library 2F). These are the items launched with the 2025/26 department-fee drive on @xmum_math (12 October 2025); paying the department fee (Freshman RM 15, Year 2 RM 10, Year 3 RM 5) unlocks the discounted prices. The merchandise to be displayed is:

Merchandise on Display	
2025 MAT T-shirt	Raglan tee: "Move, Adapt, Triumph" on the front, the MAT Student Life Cycle illustration on the back, XMUM logo and MAT logo on the sleeves. Early-bird RM 26 (department fee paid & staff) / RM 30 (other MAT students & public); normal price RM 38.
2025 MAT Tote bag	Black tote bag with the wizard-and-treasure-chest artwork (formulas spilling from the chest). Early-bird RM 12 / RM 14; normal price RM 18.
2025 MAT Keychain	Four-piece jigsaw keychain set — μ, α, τ, μ tiles that click together. Early-bird RM 5 / RM 6 per piece; normal price RM 7.
At Math Week	One sample of each item on the counter with a price card and the pre-order QR code (Google Form). Decision: whether merchandise is sold on the spot during Math Week or shown for pre-order only; stock quantities to bring; and whether a Math Week edition (e.g. a Math Week sticker or keychain tile) is added.
Pictures	 *MAT Student Council merchandise — T-shirt, keychain, tote bag (from the @xmum_math launch post, 12 Oct 2025).*

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty
MAT T-shirt (display sample)	1	From Council stock	

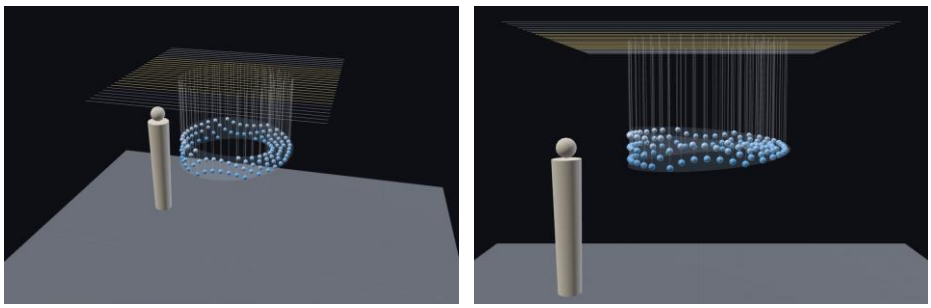
Item	Qty/Activity	Remarks	Total Qty
MAT Tote bag (display sample)	1	From Council stock	
MAT Keychain set (display sample)	1 set	From Council stock	
Price card + pre-order QR placard	1	Printed A4/A5	
Display stand / hanger for the T-shirt	1	Borrow	

Prize & Points System

Prize & Points System	
Overview	Every placard and booth feeds one set of standings: completing an activity awards points on the Leaderboard System (Math Week website), and the final standings decide the Grand Prize at the Closing Ceremony. Committee-run booths may also hand out Mini Prizes on the spot.
How points are earned	• **Code (Card) placards** — scan the QR on the placard, submit the answer on the website; correct answers score immediately, wrong answers can be retried without penalty. • **Code (Interactive) booths** — solve the physical puzzle, enter the passcode on the website. • **Digital booths** — the touchscreen game reports a win and shows a claim QR for ~25 s. • **Committee booths** — the committee member judges the win and generates a claim QR from their phone. • Each activity has its own ID: its base points can be earned **once per participant**; a repeat is acknowledged but earns nothing more.
Points scale	Decision: base points per mode — proposed Card placard 10, Digital 15, Code (Interactive) 20, Committee booth 20, plus committee-awarded bonuses (e.g. Take a Chance explanation, Rubik solver tier) equal to the base points.
MAT Stock Market	Points earned at booths are the "money" traded on the Stock Market (Days 1–6). When trading closes on Day 7 the final portfolio value is converted back into leaderboard points, so market performance feeds the same standings rather than a separate contest. Decision: conversion rate from final portfolio value to points, and whether the market can lower a participant's standing below their booth points.
Grand Prize	• Awarded to the highest-ranked participant(s) on the final leaderboard at the Closing Ceremony (Sat 24 Oct, 3.00 pm, Mini Cinema). • Equal totals (to two decimal places) share the same rank. Decision: number of Grand Prize places, and the tie policy (proposed: shared prize).
Mini Prizes	Small prizes issued on the spot by the attending committee member at committee-based booths to each participant who completes or wins the activity (e.g. Conway's Soldiers row 4, a full Knight's Tour, the Rubik solver tier, both Take a Chance explanations).
Eligibility	Open to all XMUM students and staff; participants sign in once with Student ID or email. Decision: whether organising-committee members are excluded from prizes.
Budget	RM 500 for the Grand Prize and Mini Prizes (proposal, expenditure item 6).

(Main) Installation Display

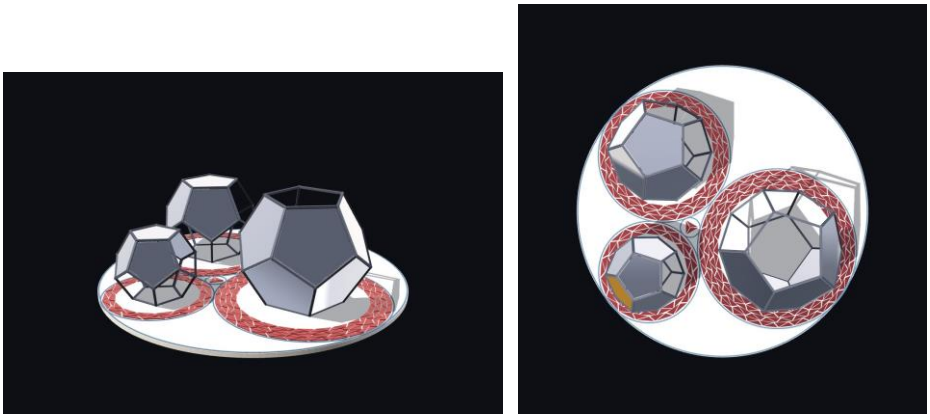
Two large installations form the visual anchor of Math Week, positioned to be visible to every visitor entering the Library. Both need a dedicated set-up window before the event (Fri 16 – Sun 18 Oct) and a dismantling window on the final day. At the Opening Ceremony the guest of honour sets the final module into the installation in place of a ribbon-cutting.

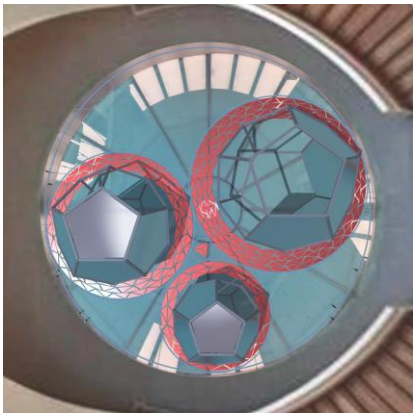
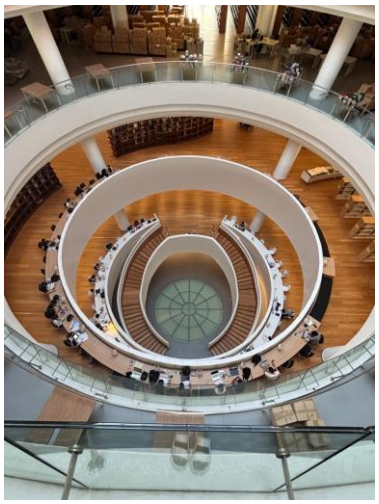
Installation Display (1) — Möbius Bead Curtain	
Concept	A hanging installation: strings drop from the Library's parallel ceiling rails and bulbs are threaded onto each string at exact heights, so that the cloud of bulbs traces a **Möbius strip** in mid-air. Seen from the right spot the bulbs read as a solid twisted band; walking round it, the band dissolves into a bead curtain.
Mathematics	The Möbius strip — a surface with one side and one edge — parametrised as P(u, v) = ((R + v cos(u/2)) cos u, (R + v cos(u/2)) sin u, v sin(u/2)). Two presets are modelled: the **Ring** (classic round strip, Ø ≈ 1.5 m) and the **Infinity** (the band follows a figure-eight standing upright facing the front — a lemniscate of Gerono — and still carries one half-twist, so it is a genuine Möbius band). Bulb heights are computed by intersecting the surface with each rail plane, so every bulb lies exactly on the surface and on a rail.
Site & dimensions	• Ceiling rails 2.6 m above the floor, parallel, 9 cm apart; a string can sit anywhere along a rail but only on a rail. • Overall diameter 1.5 m; strip centred one-third of the way down from the rails (adjustable). • Ring preset: 15 rails, 98 strings, 114 bulbs, ≈ 105 m of string. Infinity preset: 6 rails, 90 strings, 128 bulbs, ≈ 98 m of string. • One bulb type: big (Ø 5 cm) or small (Ø 2.5 cm). Decision: Ring or Infinity preset, and bulb size.
Construction	The model (*Installation Display 1/mobius_strip/viewer.html*) exports the per-string bulb positions to Excel — one row per string with rail number, position along the rail, cut length and the height of every bulb below the rail — and the geometry to OBJ. Strings are cut and knotted to the exported lengths, bulbs threaded at the listed heights, then hung on the rails at the listed positions.
Model views	 *Left: Ring preset. Right: Infinity (sketch) preset. Blue = bulbs, yellow = rails carrying strings; figure for scale.*

Installation Display (1) — Möbius Bead Curtain	
Reference	 *Reference installation style (from the earlier draft of this plan).*

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty
String / nylon cord	≈ 110 m	Ring ≈ 105 m, Infinity ≈ 98 m, plus knots	
Bulbs (beads), Ø 5 cm or Ø 2.5 cm	≈ 130	114 (Ring) or 128 (Infinity) + spares; size to be decided	
Rail hooks / clips	≈ 100	One per string	
Printed string schedule (Excel export)	1	From the model's "Save bulb positions → Excel"	

Installation Display (2) — Mirror Dodecahedra on Soddy Circles	
Concept	Three regular **dodecahedra**, each a hollow edge frame with mirror panes set into some of its faces, standing in their own circles on a round base. The offcuts of the mirror sheet are tiled as a red mosaic ring that draws each circle. Visitors look into the open faces and see the room and themselves reflected between the panes.
Mathematics	• The three circles are a genuine **Soddy configuration**: by Descartes' circle theorem $(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2)$, fixing two curvatures forces the third so that all three circles are mutually tangent and tangent to the enclosing circle; the second root of the equation gives the small **inner Soddy circle** in the gap, which is inked and tiled too. • The solids are regular dodecahedra (12 pentagonal faces, 30 edges, 20 vertices); the mirror panes turn the open frames into partial kaleidoscopes. • Every mosaic tile is an **obtuse triangle** laid in herringbone rows — the build checks each tile's largest angle exceeds 90°.
Dimensions	Table-top build (base disc Ø 600 mm, 8 mm thick): large solid edge 85 mm / height 190 mm, medium 71 / 159 mm, small 54 / 121 mm; 14 mirror panes from 2 mm sheet; 4 mm struts along all 30 edges of each solid. Sited on the Library's circular glass floor (Innovation Space 1F, radius 4.63 m) with the outer Soddy circle matched to the glass, the solids scale to 2.94 / 2.45 / 1.86 m tall (edges 132 / 110 / 84 cm). Decision: build at table-top scale, or at the full glass-floor scale with the glass as the base.
Construction	Generated by *Installation Display 2/mirror_dodecahedra/build_model.py* (OBJ/MTL for print prep and cut files, plus the viewer data). Struts run a half-width past each vertex so the corners close; each pane is pulled back 3 mm from its edges to clear the strut and sits flush with the face plane; the scrap ring is cut as zigzag strips of obtuse triangles, shrunk to 84 % for grout, from 1.5 mm sheet. Which faces carry mirror is read off the sketch's top view and is a list at the top of the script.
Model views	 *Left: side view. Right: top view — the three Soddy circles, the enclosing circle and the inner circle.*

Installation Display (2) — Mirror Dodecahedra on Soddy Circles	
In the Innovation Space	  *Left: the display placed on the glass floor (from the virtual-tour panorama and the 3F aerial photo). Right: the glass floor seen from 3F.*

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty
CIRCLE — the base and the five inked circles			
The Library's circular glass floor	—	Measured Ø 654.8 cm (r 327.4 cm). The glass IS the base — no board is built — and the outer Descartes circle is matched to it. Note ≈ 20 cm of that radius is the metal frame band rather than glass; pull the outer circle in to r ≈ 307 cm if it must sit on glass proper.	site
Navy vinyl / floor-marking tape, ≈ 24 mm	50 m	The five inked circles: outer 20.6 m + large 11.2 + medium 9.4 + small 7.1 + inner Soddy 1.4 = 49.6 m of line.	3 × 20 m rolls
SCRAPS — the red mosaic that draws each circle			
Red transparent self-adhesive film	8 m²	Ring area 9.31 m²; tiles shrink to 84 % for grout = 6.57 m² of film, plus cutting waste. ≈ 457 obtuse triangles at ≈ 349 mm base. Lazada "Transparent Red Wine Colour Glass Film" goes to 152 × 100 cm — at this volume buy by the roll and re-quote.	8 m²
Grout line	—	The 16 % gap between tiles is the grout; the film is self-adhesive, so	—

Item	Qty/Activity	Remarks	Total Qty
		nothing extra is bought. Smaller tiles multiply fast: 60 mm rows ≈ 1 500 tiles, 40 mm rows ≈ 3 400. The mosaic is the labour bottleneck.	
FRAME — the three dodecahedra			
Aluminium tube, 12–16 mm OD	76 m	30 struts × 3 solids = 69.0 m, +10 % waste. Edges 932 / 778 / 590 mm. 12 × 1.0 mm wall ≈ 6.4 kg; 16 × 1.2 mm ≈ 10.4 kg. Rigid alloy is correct now — nothing is bent, so soft wire is no longer needed.	76 m
3D-printed vertex joint	60	20 vertices × 3 solids. ALL 60 ARE THE SAME PART: the dodecahedron's vertex angles are identical at every scale and across all three solids, so one design is printed sixty times. Socket bore must match the tube. Design to follow.	60
Filament, PLA or PETG	≈ 3 kg	Sixty joints, three sockets each, plus failed prints. Confirm once the joint is designed.	≈ 3 kg
Rubber feet / pads	60	Under each solid's bottom face and anywhere tube meets glass — protects the glass floor.	60
MIRROR — the reflective faces			
Silver mirror film	17 m²	14 panes: 5 large at 1.49 m², 4 medium at 1.04 m², 5 small at 0.60 m² = 14.64 m², plus waste. Buy by the roll and re-quote at this volume — this is now the largest single material line.	17 m²
Pane substrate	14	OPEN DECISION. Film alone will not stay flat across a 932 mm pentagon — it needs either a rigid backing (2 mm acrylic or PET behind each pane) or a tensioned membrane frame. Settle this before ordering the film, because it changes both the film type and the weight.	14
TOOLS & CONSUMABLES			

Item	Qty/Activity	Remarks	Total Qty
Tube cutter, deburrer, steel rule, marker, cutting mat, squeegee, adhesive	1 set	No wire bender, no heavy-duty cutter and no ferrules — the printed joint removes all of that.	1 set

Activity Summary

Exhibition	
Expected Number • Mini Canvas Printing: • Large Canvas Printing: • Physical Models:	
Mathematics in XMU/XMUM	Mathematical Legends of XMU • Geometry of XMUM • Mathematics in the Library •
Mathematics in Daily Life	Mathematical across Culture • Malaysia • China • Indonesia • Kazakhstan Mathematics in Art • Fractal wall?
Mathematics in Curriculum/Math	• {list out the topics} •

MAT Mini Booths		
Expected Number Code (Card) Based: 20+ Digital Based: 7 Code (Interactive) Based: 5; Committee Based: 5		
Leaderboard System Digital point-recording system (email for verification) MAT Stock Market		
Mathematics Opinion Wall		• Ranking Proof Methods from Goat to Trash • Which Math Notation Annoys You the Most • If Math Were a Person
Theme 1	**Code (Card) Based**	• Age Difference Puzzle • Two-Child Probability Puzzle • Six-Matchstick Triangle Challenge

		• Fake Coin Balance Puzzle • Monty Hall Problem
	Code (Interactive) Based	• Kanoodle • Probability — Guess the Beans
	Digital Based	• Pattern Sequence • Guess 2/3 of the Average • Deal or No Deal
	Committee Based	• Conway's Soldiers • Take a Chance (Pig, Two for One, Kruskal Count, Parrondo's Paradox)
Theme 2	**Code (Card) Based**	• Cipher / Code Breaking • Passcode Collision / Birthday Problem • Mean vs Median • Shortest Path / Dijkstra's Algorithm • Equation to 45 Challenge • Target 24 Challenge
	Code (Interactive) Based	• Tower of Hanoi
	Digital Based	• Think Like KNN! — Escape Chase • Conway Infinite Loop Challenge
	Committee Based	• The Knight's Tour
Theme 3	**Code (Card) Based**	• Symmetry & Mirror Writing Challenge • Palindrome Pattern Problem • Medical Test / False Positive Probability • Mondrian Art Problem • Cryptogram Puzzle
	Code (Interactive) Based	• Bagua: The Missing Order
	Digital Based	• Symmetry Challenge
	Committee Based	• A Blind, A Deaf, A Mute
Theme 4	**Code (Card) Based**	• Pick's Theorem Area Puzzle • Falling Tree Geometry Problem • Nested Radical Challenge • Mathematical Pattern / Sum Challenge • Find the Fallacy

	Code (Interactive) Based	• Liar Dialogue
	Digital Based	• Planarity Challenge
	Committee Based	• Rubik Mini Booth (with the Rubik Society)

MAT Mini Event	
Movie Night	Good Will Hunting — Sat 24 Oct, 1.00–3.00 pm, Mini Cinema (PIC: Lee Shuan Fung)
MAT Expository Talk	Dr. Darren Ong — mathematics in patterns and symmetry in daily life; Wed 21 Oct, 7.00–9.00 pm, Library 1F (PIC: Jason William Teja)
Library Data Talk	Kow Chun Xiang — the Library data behind the Stock Market indices; Mon 19 Oct, 2.30–3.00 pm, Library 1F
Rubik Club Mini Booth	Collaboration with the Rubik Society — demos, tutorials and timed-solve challenge; Tue 20 Oct, 7.00–9.00 pm, Library 1F (contact: Cher Ming Sian)

Activity Details

Math Week Website

Math Week Website (Touchscreen Hub, Games & Leaderboard)	
Overview	The Math Week Website is the main touchscreen display: a home hub that auto-rotates between the day's highlights and four subpages, each reachable via the bottom nav or its live-preview card. • Live site: https://math-week-leaderboard.vercel.app/ • GitHub repository: https://github.com/chen18158308599-svg/Math-Week-Leaderboard.git
Home Hub	• Header: MAT logo + wordmark + dates • Panel rotates every ~20 s: today's theme/poster → Digital Based → Stock Market → Leaderboard, then loops • 3 side cards with live previews; tap to jump • Bottom nav reaches every subpage • Still needed: sponsor logos, daily poster/video content
Subpage 1 — Event Directory	• Themes by day-block, floor plan/zones, full activity list • Still needed: Math Week schedule (for each activity: name, location, time, content, pictures if possible)
Subpage 2 — Digital Based	• Actual playable game, one per day across the week; no sign-in needed to play • Win → QR → sign in once on own phone to claim; 3 attempts per puzzle, then locked • Still needed: real embed URLs from the games team
Subpage 3 — MAT Stock Market	• "Coming soon" placeholder; full specification is in the MAT Stock Market section below • Still needed: stock market website content
Subpage 4 — Leaderboard	• Live individual ranking, top 8, real-time • Still needed: nothing — fully working

Point-Claiming Flow (how wins reach the leaderboard)	
Digital games	Each digital game runs unsupervised on its own station (computer). On a win, the game reports it automatically and a QR code appears on the same screen for ~25 seconds. The student scans it with their own phone, signs in if needed, and the points are awarded instantly.
Committee-run booths	No computer needed. The committee member running the booth judges the win, then generates a claim QR on the spot from their own phone; the student scans that instead. Same points system, same leaderboard.
Card puzzles	Each printed card carries its own QR code. Scanning it opens an answer form directly: students sign in, type their answer, and correct answers score immediately. Wrong answers can be retried with no penalty.

Exhibition

The Exhibition is the in-depth half of Math Week: the mini-booths pull the foot traffic, the Exhibition is what the traffic is pulled into. It runs all week during Library opening hours and is largely self-guided. Zones (I), (II), (III) and (IV) are on Library 2F; (I), (V) and (II) are on 1F — see Floor Plan above. Every item is listed one by one under its label in Math Week Exhibition Summary.xlsx and costed in Math Week Material List.xlsx; what follows is only the main objects in each part.

Exhibition — at a glance	
The zones	• **I** — the book wall, the large display cards, the physical models and the game cards. • **II** — the string hyperboloid and the linearization artwork. • **III** — floor space allocated, nothing assigned to it yet. • **IV** — Mathematics in XMU/XMUM, and the theorem card walls. • **V** — combinatorics cards and the Clay Millennium Problems.
Still to place	*Note: include something about 67 and other meme-related content.*

Zone I.A — Mathematics on the Shelf (the book wall)	
On show	• 21 recommended books, one per bay on the Library's 6 × 6 face-out wall, in six theme columns (Mathematics for Everyone · Patterns, Networks & the Real World · Geometry & Topology · Calculus & the Classics · Algebra & Algebraic Geometry · Number Theory). Each card carries the recommender's own words. • **I.A.Q** — nine quote cards, A5 face-out. • **I.A.M** — eight model-and-art ideas from the department survey; submitted, not yet built.
Note	10 of the 21 titles are not in the Library catalogue — buy, borrow privately or drop.

Zone I.B — The large display cards	
On show	Nine A1 portrait cards: Dandelin spheres and the conics (I.B.0), Dr Darren Ong's orangutan tiling (I.B.1), wallpaper across cultures (I.B.2), mathematics in ancient China (I.B.3), Rubik and symmetry (I.B.4), the Euler characteristic (I.B.5), origami constructions (I.B.6), and kolam (I.B.7, I.B.8).
Note	Eight are finished; I.B.1 is reserved and awaiting Darren's brief.

Zone I.C — The objects	
On show	• **I.C.0** hyperbola clock — a straight rod riding the curved edge of an hourglass plate. • **I.C.1** congkak and **I.C.2** togyzkumalak, both borrowed. • **I.C.3** five ancient-China mathematics books, borrowed from the Library.

Zone I.C — The objects	
	• **I.C.4** Rubik display and **I.C.5** Rubik wall, from the Rubik Club. • **I.C.6** 3D prints and origami — 29 items (polyhedra and 4D, surfaces and knots, fractals and minimal surfaces, and the mathematics of paper folding). • **I.C.7** Sugihara optical illusions — 14 items, 10 printed and 4 paper nets. • **I.C.8** Archimedean solids from reflections — two mirror kaleidoscopes, seven solids each.

Zone I.D — The game cards	
On show	Four A4 cards standing beside the two boards: congkak (I.D.1), togyzkumalak (I.D.2) and the mathematics behind both (I.D.3a, I.D.3b) — a linear capture test driving a non-linear move. Plus I.D.4, the Rubik Soc display card.

Zone I.E — Stereographic projection	
On show	Three Segerman shadow pieces, each lit by its own point lamp so the shadow is the stereographic image of the pattern: the square-grid sphere (I.E.1), the hyperbolic hemisphere with the (7,3,2) triangle tiling (I.E.2), and the stereographic projection globe (I.E.3).
Note	All three models are downloaded, but the zone has no rows in the Summary workbook yet.

Zone II — The string hyperboloid and the linearization artwork	
On show	• **II.1** Large-scale hyperboloid of one sheet, in strings — a base ring on the floor and anchors on the ceiling rails joined by straight taut strings, so a curved surface is built out of nothing but straight lines. 260 cm tall, Ø 200 cm ring. • **Linearization artwork** — a vector field drawn as a mosaic of local affine approximations: each tile shows the field's first-order Taylor picture about its own centre, and the blank margins between tiles are exactly where the nonlinearity lives. Printed from the generator in Ref/linearization_artwork/ (needs a label and a print size).
Note	One ruling family is not self-bracing — the base ring must be pegged or taped down, not just ballasted. A Ø 200 cm ring on the floor is also a trip hazard.

Zone III — not yet assigned	
Status	Empty: no items on either zone III sheet.
Waiting for a home	Mathematics in Daily Life — Mathematics across Culture (20 A5 cards already built in Ref/cultural_displays) and Mathematics in Art; the

Zone III — not yet assigned	
	Library-data canvases from the Mathematics in the Library talk; and Geometry of XMUM. Zone III is the obvious home for them.

Zone IV — Mathematics in XMU/XMUM, and the theorem walls	
On show	• **IV.A** Mathematics in XMU/XMUM — Chen Jingrun, Goldbach, and the link to XMUM. Notes only so far. • **IV.B** the theorem card wall — 10 cards placed on the grid, 27 further titles still unplaced. • **IV.C** applied and miscellaneous — 11 cards (Zipf, RSA, FFT, image compression, Smith chart, sampling, square–cube law, phyllotaxis, Tonnetz, origami, Voronoi). • **IV.D** dynamical systems — 6 cards (logistic map, double pendulum, Sierpiński, Koch, Sharkovsky, Mandelbrot).
Note	Titles only — no copy, figures or card format chosen yet.

Zone V — Combinatorics and the Millennium Problems	
On show	• **V.A** 12 combinatorics cards — 52!, Ramsey, pigeonhole, Sperner, Catalan and the handshake problem, matching hats, Burnside, stable roommates, grid paths, Helly, Szemerédi–Trotter, Königsberg. • **V.B** the seven Clay Millennium Problems, each with a "why it matters" card — 20 card slots in all.
Note	Königsberg appears both here (V.A) and on the IV.B wall — pick one.
Group Theory & The Rubik's Cube	
Overview	A hands-on display using twisty puzzles to make group theory tangible: non-commutativity, cyclic subgroups, conjugation and parity, moving from the 3×3×3 to the 4×4×4 and a wider collection of shape modifications.
3×3×3 Standard Cube	Demonstrates: 1. **Non-abelian group operation** — the operation RU ≠ UR (not commutative). 2. **Cyclic subgroups** — repeating the sequence RUR′U′ six times returns to the start, forming a cyclic subgroup. 3. **Conjugates** — the formula ghg^{-1}, where the T-perm acts as h; this is the core of the standard blindfolded solving method. 4. **Unsolvable states** — explain why twisting a single corner creates a state that cannot be solved.
4×4×4 Standard Cube	Extends the 3×3×3 concepts to a larger puzzle and introduces new structural properties: 1. The group is much larger and gains new inner-slice moves (r, l, u, d). 2. The 4×4×4 has 24 centre pieces: swapping two same-colour centres alters the underlying group state even though the cube looks unchanged.

Group Theory & The Rubik's Cube	
	3. Flipping a single edge is impossible on a 3×3×3 but legal on a 4×4×4 — it requires an odd permutation of the inner-slice pieces. 4. Swapping just two edges is likewise legal on a 4×4×4 (but not a 3×3×3), accompanied by an unseen swap of two identical centre pieces. 5. **Reduction method** — solve the centres, then pair up the 12 edge pairs. Restricting to outer-face moves (U, R, F, B, L, D) then generates a subgroup isomorphic to the 3×3×3 cube group, so the 4×4×4 can be finished with 3×3×3 algorithms (a parity case occasionally breaks the isomorphism).
Display collection (for display only ☺)	• A timer and a speed cube for visitors to try (possible leaderboard or prize for solving within a time limit). • Standard cubes from 2×2 up to 8×8. • Mirror cube — same algebraic structure as the standard 3×3, different visual representation.  • Shape modifications: Windmill cube, Axis cube, Fisher cube, Mastermorphix. • Other mechanisms: Dino cube, Ivy cube, Molecube, Skewb, Square-1, Labyrinth, Megaminx, Pyraminx, and so on.

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty
Mirror cube	1	Don't have one — can borrow from anyone?	1

MAT Mini Booth

Mathematics Opinion Wall	
Ranking Proof Methods from Goat to Trash	Display several proof methods and invite participants to rank them based on personal preference. Participants place a sticker next to the proof method they consider the best. Design: Canva board 1
Which Math Notation Annoys You the Most	Display a selection of commonly used mathematical notations with humorous descriptions. Participants place a sticker on the notation they find the most annoying. Design: Canva board 2
If Math Were a Person	Invite participants to imagine mathematics as a person and describe its personality in one sentence. Responses are written on sticky notes and displayed on the board for others to read. Design: Canva board 3

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty
Stickers	2	For boards 1 and 2	4
Sticky notes	1	For board 3	1
Foam board	1		3
Aluminium Stand	1	Borrow from Library	3

Code (Card) Based	
Overview	Each theme has a set of printed puzzle cards. Participants solve the cards and record their answers; the collected answers are later used to decode the password for the **Mystery Chest**. Card set: MathWeek_Code_Card_Based.pdf
Theme 1	General mathematics puzzle cards; answers recorded on an answer sheet.
Theme 2	Theme-related mathematics puzzle cards; answers contribute to the final Mystery Chest decoding challenge.
Theme 3	Puzzles on symmetry, patterns, arts and logical reasoning; answers recorded for the final decoding challenge.
Theme 4	Mathematics- and science-related puzzle cards involving mathematical reasoning and applications; answers recorded for the final decoding challenge.

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty

Digital Based	
Pattern Sequence (Theme 1)	Participants identify the underlying pattern in a sequence and predict the next element. Tests logical reasoning and pattern recognition.
Guess 2/3 of the Average (Theme 1)	Participants choose a number and try to predict two-thirds of the average of all submitted numbers. Demonstrates strategic thinking and basic game theory.
Deal or No Deal (Theme 1)	Participants decide between accepting an offer or continuing the game based on the remaining possible rewards. Introduces probability, expected value and risk.
Think Like KNN! — Escape Chase (Theme 2)	Participants select the K nearest data points and predict the class by majority vote. Their performance determines how far the character progresses in the escape challenge.
Conway Infinite Loop Challenge (Theme 2)	Participants interact with Conway's Game of Life and attempt to create a repeating or stable pattern by choosing an initial configuration.
Symmetry Challenge (Theme 3)	Participants complete visual symmetry and reflection challenges by identifying or reconstructing the mirrored pattern.
Planarity Challenge (Theme 4)	Participants rearrange the vertices of a graph so that no edges intersect, introducing planar graphs in graph theory.

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty

Code (Interactive) Based	
Kanoodle (Theme 1)	
Setup	A large-scale floor version of Kanoodle built from **10 × 10 cm EVA foam cubes**. The board is the standard 5 × 11 grid (50 cm × 110 cm), marked out with tape on a base board or the floor. The 12 pieces are built by fixing cubes together (55 cubes in total); the Kanoodle letter of each piece (A–L) is written on the top of every cube of that piece. The full piece list is in the **Kanoodle piece list** table below. A puzzle card at the station shows the starting position — a **hard** puzzle with only two pieces pre-placed and ten to place — and marks five cells of the board with a star (★). Participants place the pre-placed pieces first, then solve. The card (starting position and solution) is the **Kanoodle puzzle card** table below. • Reference videos of harder starting positions: example 1, example 2.
Question	Starting from the position on the card, fill the whole 5 × 11 board using all 12 pieces — no overlaps, no gaps and no leftover pieces. Then read off the letters of the pieces covering the five ★ cells, in reading order (top row first, left to right).
Solution	The passcode is the 5-letter string of piece letters on the ★ cells: **CLBHG** for Puzzle Card 1. The completion of the starting position is unique, so every solver reads the same passcode. Participants enter it on the Math Week website to claim points.
Tower of Hanoi (Theme 2)	
Setup	A purchased, ordinary mid-size Tower of Hanoi set (three pegs labelled A, B, C from left to right; disks numbered 1 = smallest). The challenge uses **4 disks**, starting on peg A with target peg C. Rules: move one disk at a time; a disk may never be placed on a smaller disk. The optimal solution ($2^4 - 1 = 15$ moves) is unique, so every move of it is determined — this is what makes the question checkable.
Question	Move the whole tower from peg A to peg C in the minimum number of moves, counting your moves. In the optimal solution, which disk is moved on the **5th move**, and onto which peg?
Solution	The passcode is the disk number followed by the destination peg. The optimal 4-disk solution begins 1→B, 2→C, 1→C, 3→B, **1→A**, so the 5th move is disk 1 onto peg A and the passcode is **1A**. Committee check without playing it out: on move k the disk moved is 1 + (number of trailing zeros in the binary form of k) — k = 5 = 101_2 has none, so disk 1 moves. With an even number of disks, disk 1 circulates A → B → C → A, so its 3rd move (move 5) lands on A.
Bagua: The Missing Order (Theme 3)	
Setup	A whiteboard with eight movable trigrams (two layers of cardboard with a magnet inside), a marked starting position, and a printed Taiji symbol as decoration. Each trigram is read as a 3-bit binary number: **solid line = 1, broken line = 0**, so the eight trigrams are the numbers 0–7 (000 = 0 …

Code (Interactive) Based	
	111 = 7). A stack of nine numbered hint cards (Easy 1–4, Medium 1–3, Hard 1–2) sits at the station; the participant draws one card. Hint types on the cards: **pattern hints** (relationships or positions of selected trigrams, often via the binary number system) and **memory hints** (the correct order is shown for a few seconds, then rebuilt from memory). The nine problems are listed in the **Bagua problem set** table below. Decision: memory-hint cards need a committee member to show the order for a few seconds — decide whether to keep them at an unsupervised station, and whether a third hint type is added.
Question	Using the clues on the card, arrange the eight trigrams in the correct order along the board, starting from the marked position.
Solution	The passcode is the final order written as eight decimal digits, one per trigram (0–7). For example the order 111, 110, 101, 100, 011, 010, 001, 000 gives **76543210**. Every card has its own 8-digit passcode; the answer key is kept by the committee and loaded into the Math Week website.
Liar Dialogue (Theme 4)	
Setup	A printed dialogue card with two characters: **Nailong**, who always tells the truth, and **Niulai**, who always lies. The card shows a short conversation between them about a hidden number (or hidden fact), in the style of a knights-and-knaves puzzle. Decision: the dialogue content still has to be written — number of cards, the hidden values, and whether the speakers' names are hidden so participants must also work out who is lying.
Question	From the statements in the dialogue, work out what is true and deduce the hidden number (or fact).
Solution	The passcode is the hidden value deduced from the dialogue, e.g. **42**. Each card has one passcode, kept in the committee's answer key.
Others — Luban Locks & Topology Puzzles (free play, all week)	
Setup	A tray of hands-on puzzles that visitors may pick up and play with at any time: the **36-piece beech Kongming / Luban lock set** (interlocking burr puzzles, difficulty 1 and 2) and the **13 string-and-ring threading puzzles** (解绳 — topology puzzles). Each puzzle sits in a labelled slot with a photo of its assembled state, so it is obvious what "put back together" means. The full list is in the **Luban locks & topology puzzles model list** table below. The mathematics on the label: burr locks are exercises in **interlocking assembly** — the order of insertion matters and the last piece is the key; the rope puzzles are **topology** — the ring can only be freed by a continuous deformation of the cord, never by force.
Question	Take one apart, work out how it goes together, and **put it back exactly as you found it** before you leave the tray. Try the rope puzzles without untying anything.

Code (Interactive) Based	
Solution	No passcode and no points — this is play for its own sake. The one rule is that every puzzle is returned assembled. The committee checks the tray at the morning and afternoon rounds and reassembles anything left in pieces (assembly photos are kept at the tray for this).
Probability — Guess the Beans (Theme 1)	
Setup	A sealed glass jar filled with a known number N of cheap identical items (e.g. beans or beads), counted by the committee beforehand and kept secret. A guess sheet or QR form is placed beside the jar. The display illustrates the **law of large numbers** / wisdom of the crowd: the average of many independent guesses tends to be close to N. A second poster shows the running average of all guesses so far; a participant who has seen it and guesses again is no longer independent (an anchoring demonstration). Decision: the item to use, the value of N, the tolerance band for scoring, and whether the second (anchored) guess is displayed as a demonstration only or also scored.
Question	(1) How many items are in the jar? (2) After seeing the running average of everyone's guesses, would you change your guess?
Solution	The passcode is **N**, the exact count. On the Math Week website a first guess within the tolerance band of N scores; only the first (independent) guess counts for points.

Supporting detail **— Kanoodle — Piece List (12 pieces, 55 cubes)**			
Piece	**Polyomino**	**Cubes**	**How to build**
A	L-tetromino	4	3 cubes in a row, plus 1 cube beside the last one (an L).
B	P-pentomino	5	A 2 × 2 square, plus 1 cube extending one side (a 2 × 3 block with one corner missing).
C	L-pentomino	5	4 cubes in a row, plus 1 cube beside the last one.
D	Y-pentomino	5	4 cubes in a row, plus 1 cube beside the second one.
E	N-pentomino	5	2 cubes in a row and 3 cubes in a row, joined side by side and offset by one (an S/N shape).
F	L-tromino	3	2 cubes in a row, plus 1 cube beside the end (a small corner).
G	V-pentomino	5	3 cubes in a row and 3 cubes in a column sharing the corner cube (a big L).
H	W-pentomino	5	A staircase: 1 cube, then 2 cubes stepped down, then 2 cubes stepped down again.
I	U-pentomino	5	3 cubes in a row, plus 1 cube on each end on the same side (a U).

Supporting detail — **Kanoodle — Piece List (12 pieces, 55 cubes)**			
Piece	**Polyomino**	**Cubes**	**How to build**
J	I-tetromino	4	4 cubes in a straight row.
K	O-tetromino	4	A 2 × 2 square.
L	X-pentomino	5	A plus sign: 1 centre cube with 1 cube on each of its four sides.
Total	1 tromino, 3 tetrominoes, 8 pentominoes	**55**	Fills the 5 × 11 board exactly. Write the piece letter on the top face of every cube.

Supporting detail — **Kanoodle — Puzzle Card 1 (hard: two pieces placed)**	
Starting position	Pre-placed pieces (shaded): **A, D**. Pieces to place: **B, C, E, F, G, H, I, J, K, L**. The five ★ cells give the passcode, read in reading order (top row first, left to right).
Solution	The completion is unique (checked by computer search). ★ cells are shaded yellow.

					D	★				
		★			D	A	A	A	★	
			★	D	D	A				
					D					★

I	I	L	C	C	C	C	K	K	B	B
I	L	L	L	H	D	C	K	K	B	B
I	I	L	H	H	D	A	A	A	B	G
F	F	H	H	D	D	A	E	E	E	G
F	J	J	J	J	D	E	E	G	G	G

Passcode: **CLBHG**

Supporting detail — **Bagua: The Missing Order — Problem Set (9 problems)**	
Easy 1	Sequence 111, 110, ?, ?, 001, 010, 001+001, 111−010 — the 3rd, 4th, 7th and 8th trigrams are missing. The 3rd and 4th are IQ-test-style pattern recognition; the 7th and 8th are simple binary calculations.
Easy 2	Arrange the trigrams in ascending binary order: 000, 001, 010, 011, ….
Easy 3	Each trigram is shown beside a basic arithmetic problem whose answer is a number from 1 to 8; no two trigrams share a number. Place each trigram at the position given by its answer.

Supporting detail — **Bagua: The Missing Order — Problem Set (9 problems)**	
Easy 4	Some positions are given (e.g. the 5th is 110, the 8th is 011) and the rest are deduced logically (e.g. "the 1st has exactly two solid lines", so it is 101). Decision: how many trigrams are left missing on this card.
Medium 1	Binary calculations shown as pictures, e.g. the first position is (011 + 010) × 100. Participants convert the pictures to binary numbers and compute the result.
Medium 2	System of equations in 2 variables: 6 positions are known, 2 are unknown. E.g. let 110 = x and 101 = y with x + y = 3, x − y = −1, giving x = 1, y = 2, so 110 is 1st and 101 is 2nd.
Medium 3	Same format as Easy 3, but with harder mathematics problems (number theory may be involved).
Hard 1	System of equations in 3 variables: 5 positions are known and 3 are missing, in the same style as Medium 2.
Hard 2	Identify the correct trigram from several clues about the number and position of solid/broken lines, e.g.: • It has exactly 1 broken line • The broken line is not at the top • The bottom line is broken • The top line is solid

Supporting detail — **Luban Locks & Topology Puzzles — Model List (36-piece set + 13 rope puzzles)**

A. Kongming / Luban locks (鲁班锁 · 孔明锁) — interlocking burr puzzles, 36-piece beech set

No.	Model	中文	Difficulty · size
A1	Six-way lock, 6-piece burr	六通锁	难度一 · 7.5 cm
A2	Three-way lock, 3-piece burr	三通锁	难度一 · 8.8 cm
A3	Hero lock	好汉锁	难度一 · 7.5 cm
A4	Chinese cross	十字架锁	难度一 · 10 cm
A5	Six-plate assembly	六片拼	难度一 · 7.5 cm
A6	Nine-way lock, little pineapple	九通锁	难度一 · 7.5 cm
A7	Wall-corner joint	墙角	难度一 · 6 cm
A8	Segmented Luban sphere	鲁班球	难度一 · 7.5 cm
A9	Pyramid lock (binary / ternary)	金字塔鲁班锁	难度一 · 7 cm
A10	Ball in cage	笼中取珠 / 笼中取宝	难度一 · 7.5 cm, two pieces
A11	Fifteen-way lock	十五通	难度一 · 7.5 cm
A12	中-character lock	中字锁	难度一 · 7.5 cm

Supporting detail — **Luban Locks & Topology Puzzles — Model List (36-piece set + 13 rope puzzles)**			
A13	米-character / plum-blossom lock	米字锁 · 梅花锁	难度一 · 7.5 cm
A14	Look-back lock	回首	难度一 · 7.5 cm
A15	Nine-bar plain	九根	难度一 · 7.5 cm
A16	Twenty-four lock, lattice frame	二十四锁	难度一 · 7.5 cm
A17	井-character lock, 12-piece	井字锁	难度一 · 7.5 cm
A18	Besieged city	围城	难度二 · 10 cm
A19	Dragon-tail snake lock	神龙摆尾	难度二 · 6 cm
A20	Crosshair	准心	难度二 · 7.5 cm
A21	Eight-point star ball	八角球	难度二 · 7.5 cm
A22	Fourteen-bar burr	十四根	难度二 · 7.5 cm
A23	Twin burr, brothers of one heart	兄弟同心	难度二 · 7 cm
A24	Cube lock	正方锁	难度二 · 6 cm
A25	Round Luban ball	圆球鲁班锁	难度二 · 6 cm
A26	Football ball	足球	难度二 · 7.5 cm
A27	Twin hearts	心心相印	难度二 · 7.5 cm
A28	Eighteen Arhats, 18-piece burr	十八罗汉	难度二 · 7.5 cm
A29	Heart lock	心锁	难度二 · 6 cm
A30	Draw-together lock	靠拢 (十二兄弟)	难度二 · 7.5 cm
A31	Big pineapple, 33-piece burr	大菠萝	难度二 · 10 cm
A32	Kongming ball	孔明球	难度二 · 7.5 cm
A33	Double-cross cage	双十字笼	难度二 · 7.5 cm
A34	Wine-barrel lock	酒桶锁	难度二 · 7.5 cm
B. String-and-ring / threading rope puzzles (解绳 · 解扣) — topology			
No.	**Model**	**中文**	**Notes**
B1	Mouse in ladder-rings (ring staircase)	鼠困梯环	free the ring without untying
B2	Golden-rooster post: slotted post + loose ring + bead-post	金鸡立柱	free the ring without untying
B3	Double-post rope: two slotted posts + loose ring	双柱解绳	free the ring without untying
B4	Single-post rope: slotted post + ring + two discs	单柱解绳	free the ring without untying

Supporting detail — **Luban Locks & Topology Puzzles — Model List (36-piece set + 13 rope puzzles)**			
B5	Three bows in the peach garden: tripod + two key-rings	桃园三拜	free the ring without untying
B6	Escape from certain death: tall slot with ring through it	死里逃生	free the ring without untying
B7	Big head: disc with holes	大头	free the ring without untying
B8	Cute: block with two cords	呆萌	free the ring without untying
B9	Robot: two uprights joined	机器人	free the ring without untying
B10	Triangle Q: triangular frame	三角 Q	free the ring without untying
B11	3D triangle: pyramid with through-hole	立体三角	free the ring without untying
B12	Alloy seahorse	合金海马解绳	free the ring without untying
B13	Magic ring escape, hand-held paddle (sold separately)	魔环逃生	free the ring without untying
The set lists 36 pieces: A10 (ball in cage) is two pieces and A13 covers the 米字 and 梅花 names of one item, so 34 rows. Photos of every assembled state are printed for the tray.			

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty
Kanoodle (Theme 1)			
EVA foam cubes, 10 × 10 cm	55 + spares	To build the 12 pieces (see piece list)	
Strong glue / double-sided tape	TBD	To join the cubes into pieces	
Permanent marker	2	To write the piece letters on the cubes	
Floor tape or base board (50 × 110 cm)	1	To mark the 5 × 11 grid and the ★ cells	
Puzzle card (starting position)	1	Printed from the puzzle card table; hard (two pieces pre-placed)	
Tower of Hanoi (Theme 2)			
Tower of Hanoi set, ordinary mid-size	1	Purchased; 4 disks used for the challenge	
Bagua: The Missing Order (Theme 3)			
Cardboard	TBD	For the trigrams and the base	
Blank cards	2 sets	Shopee link — shared with Blind, Deaf, Mute	

Item	Qty/Activity	Remarks	Total Qty
Whiteboard	1	Borrowed on campus; the trigrams are stuck on it	
Neodymium disc magnets	30 pcs	Diameter 10 mm, thickness 2 mm	
Double-sided tape	1	To stick the magnets onto the trigrams	
Marker pens	TBD	For writing hints on the blank cards	
Mathematics symbol stickers (optional)		To decorate the blank cards	
Printed Taiji symbol	1	Printed on A4 paper	
Scissors or blade (刀片)	1	To cut the cardboard	
Liar Dialogue (Theme 4)			
Printed dialogue cards	TBD	One passcode each	
Probability — Guess the Beans (Theme 1)			
Glass jar with lid	1		
Beans / beads (cheap identical items)	N TBD	Counted beforehand	
Guess sheet or QR form + poster for the running average	1		
Others — Luban Locks & Topology Puzzles			
36-piece beech Kongming / Luban lock set (榉木全套)	1 set	Taobao — Material List, Purchase sheet (Urgent)	
String-and-ring threading puzzle set (13 designs)	1 set	Taobao — Material List, Purchase sheet (Urgent); 魔环逃生 sold separately	
Tray with labelled slots + assembled-state photos	1	Printed labels with the puzzle name and its mathematics	

MAT Mystery Puzzle	

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty

Committee Based	
Conway's Soldiers (Theme 1)	
Setup	A grid board with a horizontal line across it and a supply of counters ("soldiers"). Soldiers start on cells below the line. One rule: a soldier jumps horizontally or vertically over an adjacent soldier onto an empty cell, and the jumped-over soldier is removed (as in checkers). The booth uses a **custom grid mat** (10 columns × 8 rows, taped or printed) with the line marked between rows +1 and 0, so there are 4 rows above the line for the target and 4 rows below for the army, and a supply of 30+ counters. The online version is for the committee's reference only: https://www.cleverlearning.co.uk/blogs/blogConwayInteractive.php. Board size: the row-4 solution needs 20 soldiers in a block 7 columns wide and 4 rows deep (a standard 8 × 8 checkerboard would fit it only with the target column off-centre, and leaves no room for participants to experiment); the 10 × 8 mat also fits the 9-wide variant B. The start position and the 19-jump sequence are in the **Conway's Soldiers reference solution** table below.
Question	Place as many soldiers as you like below the line, then get one soldier to the 4th row above the line. (Bonus challenge: try for the 5th row.)
Solution	Minimum numbers of soldiers: row 1 → 2, row 2 → 4, row 3 → 8, row 4 → 20. The committee member checks the reached row and awards points for row 4. Row 5 is provably impossible (Conway's argument: weight each cell by φ^(−distance) to the target, where φ is the golden ratio; the total weight of all soldiers below the line is less than the weight of the target and can never increase).
The Knight's Tour (Theme 2)	
Setup	A standard 8 × 8 chessboard, one knight, and a set of **checkers pieces** used as markers: the committee member places a checker on every square the knight leaves, so the path is tracked physically and revisits are visible at once; each move is checked to be a legal knight move. All three variants are offered, and the participant chooses one: • **Full tour** — the whole 8 × 8 board (64 squares). • **Small board** — a 5 × 5 or 6 × 6 corner of the board, its border marked with tape. • **Holes** — an 8 × 8 or 6 × 6 board with a few squares marked as holes (paper markings) the knight may never land on. Online demo board for the committee: https://www.maths-resources.com/knights/ (alternative: enclose.horse). Decision: time limit per attempt, and whether the small-board variant earns fewer points than the full tour.
Question	Starting from any square, move the knight so that it visits every square of the board exactly once.
Solution	The committee member verifies that every square of the chosen board carries a checker and none was visited twice, then generates the claim

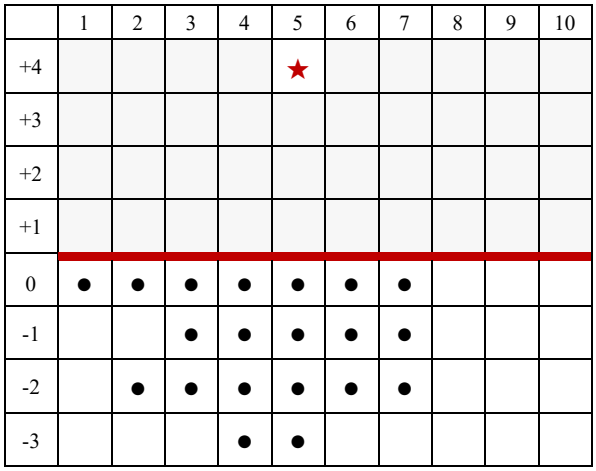
Committee Based	
	QR. Reference for the committee: **Warnsdorff's rule** — always move to the square from which the knight has the fewest onward moves — almost always produces a full tour on the 8 × 8 board; a printed reference tour is kept at the booth.
Rubik Mini Booth (Theme 4) — run with the Rubik Society	
Setup	A booth staffed by Rubik Society members next to the *Group Theory & The Rubik's Cube* exhibition, with several 3 × 3 cubes, a speed-cube timer and printed beginner method sheets. Members scramble the cubes, run the timer and teach the first-layer method to beginners. See also the Rubik Club Mini Booth entry under MAT Mini Event. Decision: the time limits for the two tiers, and whether an intermediate tier (e.g. first two layers) is added.
Question	Beginner tier: solve one full layer of a scrambled cube within the time limit, using the method sheet if needed. Solver tier: solve the whole cube within the time limit.
Solution	A society member confirms the solved layer / solved cube and the time on the timer, then awards points via the claim QR (higher points for the solver tier). Fastest times of the day can be posted at the booth.
Luban Lock Challenge (Theme 3)	
Setup	A timed challenge on one specific puzzle from the free-play tray (see **Others — Luban Locks & Topology Puzzles** under Code (Interactive) Based and its model list). The committee member keeps two copies of the challenge puzzle: one assembled as the reference, one handed over in pieces. A stopwatch and a printed assembly photo (face down, for hints) are on the table. Decision: which puzzle is the challenge — pick from the model list, e.g. A1 six-way lock (classic, moderate), A28 Eighteen Arhats or A31 big pineapple (hard), or a rope puzzle such as B1 mouse in ladder-rings; also the time limit and whether hints cost points.
Question	Assemble the puzzle from its loose pieces within the time limit (rope puzzle variant: free the ring, then thread it back).
Solution	The committee member compares the result with the reference copy — every piece in place and the lock holding together (or the ring back on its cord) — and awards points via the claim QR; a mini prize for a solve without hints. The committee member learns the disassembly / assembly sequence beforehand from the set's instruction sheet.
Take a Chance (Theme 1) — dice games and the Kruskal Count	
Setup	A probability booth with three parts, each chosen for a surprising probabilistic idea (rules in the **Take a Chance — game rules** table below): • **Dice games** — Pig (when to stop is an expected-value decision) and Two for One (which die is really faster), played against the committee member.

Committee Based	
	• **Kruskal Count** — the classical card trick based on the coupling of two random walks through a shuffled deck; the committee member has practised the trick and its explanation. • **Parrondo's Paradox** — two games that each lose money on their own, but win when they are alternated; played with percentile dice (two d10 read as 00–99) and a stack of counters as capital. Kruskal Count set-up: a shuffled standard deck is dealt face up in a row. Card values: Ace = 1, number cards = face value, Jack/Queen/King = 5. The participant secretly chooses a number from 1 to 10 and silently "walks" through the row: they look at the card in that position, its value tells them how many cards to jump forward, and so on until a jump would go past the end of the row. The last card they land on is their **key card**. The committee member does the same walk from their own secret start. **Rounds and points.** Pig: best of 3 rounds against the committee member. Two for One: best of 3 races, the participant choosing the dice first. Kruskal Count: one run. Parrondo: 30 rounds each of game A alone, game B alone and the alternating pattern ABBAB…, tracked on a score sheet. A participant earns the booth's base points once for winning either dice game, and a bonus (the same base points again) for correctly explaining either the Kruskal coupling or why the Parrondo combination wins; a mini prize on the spot for a full explanation of both.
Question	Dice: beat the committee member at Pig or Two for One. Cards: keep your start number and key card secret while the committee member names your key card — then explain why the trick (almost) always works. Parrondo: play games A and B separately and watch your counters shrink, then play them alternately — and explain why the combination wins.
Solution	Dice: the committee member judges the winner (win conditions in the game rules table). Cards: the two walks land on the same card as soon as they share one card, and from then on they move together — the chains have **coupled**. With an average jump of about 4.5 cards over a 52-card deck, the walks couple with probability roughly 85%, so the committee member's key card is (almost always) the participant's. Parrondo: game B is only favourable when your capital is not a multiple of 3; alternating with game A reshuffles the capital so that you spend more rounds in B's favourable state than B alone would allow, and the mixture drifts upwards even though each game drifts down. Points as set out under Setup.
A Blind, A Deaf, A Mute (Theme 3)	
Setup	Three participants with different restrictions: the **Blind** wears an eye mask and makes the final decision; the **Deaf** wears earmuffs and cannot hear; the **Mute** may not speak. The information needed to solve the puzzle is split between them: the Mute and the Deaf each hold a hint card, and the committee tells the Blind the remaining piece. No single

Committee Based	
	person has everything. Puzzle formats are listed in the **puzzle types** table below; reference video for puzzle design: https://youtu.be/dG07dsc8f1A?si=4lJp7hgTf0HgS3cH. The Blind is **really blindfolded** (eye mask on throughout), so their information must be given by touch or told to them. Decision: time limit per puzzle (proposed 5 minutes).
Question	Cooperate under the restrictions to solve the puzzle; within the time limit the Blind must state the final answer. Example: find f(x, y) where the Mute holds "$f(x, y) = x^2 + y^2$", the Blind holds "$x = 2, y = 3$" (tactile pieces or told by the committee) and the Deaf must relay information between them.
Solution	The committee's copy of each puzzle card carries the answer (for the example: $2^2 + 3^2 = \mathbf{13}$). A correct answer within the time limit earns points via the claim QR.

Supporting detail **— Take a Chance — Game Rules**	
Pig (dice)	Contestants take turns. On a turn, roll a die as many times as you like, adding each roll to your turn total — but rolling a 1 ends the turn immediately and wipes that turn's points; stop at any time to bank the turn total. Win: highest banked score after the agreed number of rounds. The mathematics: when to stop is an expected-value decision. Best of 3 rounds.
Two for One (dice)	Contestants choose between two standard six-sided dice or one twelve-sided die, then race to roll every number from 2 to 12. Win: first to complete the set. (The single d12 is faster on average: with two dice, 2 and 12 each appear only once in 36 rolls.)
Kruskal Count (cards)	Deck dealt face up in a row; Ace = 1, number cards = face value, J/Q/K = 5. Both players pick a secret start position (1–10) and jump forward by the value of each card they land on until they cannot jump further; the last card landed on is the key card. • The committee member announces their key card; if it matches the participant's, the trick has worked. • Explanation to give afterwards: two random walks on the same row coincide forever once they share a single card (coupling). Small average jumps make sharing a card very likely — about 85% with these values, and higher if face cards count as 1 instead of 5. • Win: the participant explains the coupling idea in their own words.
Parrondo's Paradox (percentile dice)	Capital starts at 20 counters; each round wins or loses 1 counter. Roll two d10 and read them as a number from 00 to 99. • **Game A:** win if the roll is 00–48 (probability 0.49) — a slightly losing coin. • **Game B:** if your capital is a multiple of 3, win only on 00–08 (probability 0.09); otherwise win on 00–73 (probability 0.74). On its own B is also losing.

Supporting detail — **Take a Chance — Game Rules**	
	• **Combined:** play strictly in the pattern A, B, B, A, B, B, … The combination is winning: the A rounds keep you out of B's bad "multiple of 3" state more often than B alone does. (The pattern matters — plain alternation A, B, A, B, … still loses.) • Play 30 rounds of each of A, B and the combination and compare the three final capitals. Expected drift per round (simulated): A ≈ −0.02, B ≈ −0.02, ABBAB ≈ +0.055 counters — small, so a pre-run simulation chart of 1000 rounds (A ≈ −21, B ≈ −18, ABBAB ≈ +55) is displayed at the booth to show the long-run trend. Win: the participant explains why the combination wins.

Supporting detail — **Conway's Soldiers — Row-4 Reference Solution (20 soldiers, 19 jumps)**	
Board	Custom grid mat, 10 columns × 8 rows. The line runs between rows +1 and 0 (thick red line); rows +1 to +4 are above it, rows 0 to −3 below. Cells are written (column, row). The target ★ is column 5, row +4.
Start position	Variant A of the standard 20-soldier arrangement (Berlekamp, Conway & Guy): 7 + 5 + 6 + 2 soldiers, 7 columns wide, 4 rows deep, with the target column the 3rd from the right edge of the formation.  Rows 0, −1, −2, −3: columns 1–7; columns 3–7; columns 2–7; columns 4–5.
Jump sequence	Found by computer search and verified; every jump is orthogonal over an adjacent soldier onto an empty cell. 1. (3,-1) jumps over (3,0) to (3,+1) 2. (4,-1) jumps over (4,0) to (4,+1) 3. (7,-1) jumps over (7,0) to (7,+1) 4. (6,-1) jumps over (6,0) to (6,+1) 5. (4,-3) jumps over (4,-2) to (4,-1) 6. (2,-2) jumps over (3,-2) to (4,-2) 7. (1,0) jumps over (2,0) to (3,0) 8. (3,0) jumps over (3,+1) to (3,+2) 9. (7,+1) jumps over (6,+1) to (5,+1) 10. (4,-2) jumps over (4,-1) to (4,0)

Supporting detail — **Conway's Soldiers — Row-4 Reference Solution (20 soldiers, 19 jumps)**	
	11. (5,0) jumps over (5,+1) to (5,+2) 12. (5,-2) jumps over (5,-1) to (5,0) 13. (4,0) jumps over (4,+1) to (4,+2) 14. (7,-2) jumps over (6,-2) to (5,-2) 15. (5,-3) jumps over (5,-2) to (5,-1) 16. (5,-1) jumps over (5,0) to (5,+1) 17. (5,+1) jumps over (5,+2) to (5,+3) 18. (3,+2) jumps over (4,+2) to (5,+2) 19. (5,+2) jumps over (5,+3) to (5,+4) After jump 19 a single soldier stands on (5,+4) — row 4.
Smaller targets	• Row 1: 2 soldiers in a column directly below the target. • Row 2: 4 soldiers — three in row 0 plus one below the rightmost. • Row 3: 8 soldiers — five in row 0 plus three in a column under the middle one (or two beside it). • Row 5: impossible for any finite army (golden-ratio weight argument), so it is the "Nobel prize" bonus.

Supporting detail — **A Blind, A Deaf, A Mute — Puzzle Types**	
System of equations (3 variables)	Mute and Deaf each hold a hint card carrying one equation of the system; the committee tells Blind the remaining equation, and Blind must solve for the unknowns.
Geometry	Mute knows the perimeter of a shape; Blind knows the relationship between the side lengths and is told the geometric formula; Deaf is given an angle where the problem needs one.
Logical clues	For example: A, B, C are three different integers from 1–6. A is greater than B. C is even. A + C = 7. As stated these clues allow several answers (e.g. A = 5, C = 2, B = 1 or A = 3, C = 4, B = 1). Tightened version: A, B, C are three different integers from 1–6; A is greater than B; C is even; A + C = 7; A + B + C = 10. Unique answer A = 5, B = 3, C = 2.

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty
Conway's Soldiers (Theme 1)			
Custom grid mat, 10 × 8 cells, line marked	1	Taped on card/foam board or printed A1; columns numbered 1–10, rows +4 … −3	
Counters (soldiers)	30+	Row 4 needs 20; the Knight's Tour checkers can double up	
Printed reference solution	1	From the reference solution table, for the committee only	
The Knight's Tour (Theme 2)			

Item	Qty/Activity	Remarks	Total Qty
Chess board with a knight piece	1	Standard board; smaller variants use the same board with restricted borders	
Checkers pieces (markers for visited squares)	64	Two checkers sets, or any 64 flat counters	
Paper markings for "holes" and border tape	1 set	For the holes and small-board variants	
Rubik Mini Booth (Theme 4)			
3 × 3 cubes	4+	Provided by the Rubik Society	
Speed-cube timer	1	Provided by the Rubik Society	
Printed beginner method sheets	TBD		
Take a Chance (Theme 1)			
Standard deck of cards	2	For the Kruskal Count (one in use, one spare)	
6-sided dice	2	Pig and Two for One	
12-sided die	1	Two for One; collections of x-sided dice are cheap, e.g. Shopee (Indonesia) listing	
Percentile dice (two d10, one marked as tens)	1 pair	Parrondo's Paradox	
Counters (capital)	60	Parrondo's Paradox — 20 per game	
Printed simulation chart (Parrondo, 1000 rounds)	1	A4, at the booth	
Score sheets + pen	1 set	Pig and Parrondo	
A Blind, A Deaf, A Mute (Theme 3)			
A4 paper	1 set	For participants to make notes and submit answers	
Pens	TBD		
Eye mask	2	Blind role	
Earmuffs / sleeping ear covers	2	Deaf role	
Blank cards	—	Shared with Bagua (see above)	
Luban Lock Challenge			
Challenge puzzle (model to decide)	2	One reference copy + one to solve; bought as singles (Purchase sheet lists single prices)	
Stopwatch / timer	1	Phone timer is fine	

Item	Qty/Activity	Remarks	Total Qty
Assembly photo / instruction sheet	1	From the set	

MAT Stock Market (with Digital Point-Recording System)	
Overview	A simulated stock-trading game running on the Math Week website. Participants earn "money" by playing booths, then invest it in 5 indices built from real school data. Rankings are by total portfolio value. Participants can email the committee for any verification. Technical formulas and rules are in the **Technical Specification** table below.
Joining Flow	1. No guests. Scan QR to join with Student ID or email. Four UI buttons: Leaderboard, Profile, Stock Market, History. 2. Do leaderboard users have to log in with the school email? Alternatively verify, then give a nickname (nicknames supervised by Admin in case of inappropriate names). 3. The system builds a participant_id; the profile page shows a personal QR and the stocks bought. 4. Stock Market: interact with the different stocks and see the information for each. 5. History: the record of the participant's "money".
Earning Money at Booths	1. After joining, participants go to booths and play games. 2. Digital booths: the platform records completion and marks correctness automatically. 3. Physical booths: staff scan the participant's QR and select the configured outcome. 4. History immediately shows the points added, and what was learned / progress toward theme badges? 5. Each activity has its own ID, so its base points can be earned only once; repeated scans show "already done" instead of awarding points again.
Getting Started (Market Brief)	1. On first entering the Stock Market a "Market Brief" appears — a beginner's guide explaining the stocks. It has no answers and does not affect rankings. 2. Participants compare data, then choose to buy or sell. 3. The trade confirmation page shows units, price, handling fee, remaining cash and post-trade positions. 4. Portfolio value = cash + Σ (units held × latest published price) — "现金 + Σ（持有单位×最新公布价格）".
Trading Rules (summary)	• Everyone starts with $5000 and no holdings. • Whole units only; minimum trade is 1 unit. • Long-only: no short selling, borrowing/margin, derivatives, transfers, gifts or P2P trading (没有做空、借贷/杠杆/衍生品、转让、赠送或 P2P 交易). • 1% fee per trade, minimum $1 (see formulas below).

MAT Stock Market (with Digital Point-Recording System)	
	• One confirmed action per index per day (buy, sell or hold) — at most 5 decisions a day; no buying and selling the same index on the same day. • Confirmed trades are irreversible; Admin "reversal" is used only for a discussed, justified reason. • No timer or countdown while trading.
Market Timeline	• **Days 1–6:** participants buy/sell to predict the next day's prices. Each morning the system reloads the day's information and updates prices with an explanation. • **Day 7 (25/10):** final update; trading freezes. Final net worth and the market-performance component of the event scores are calculated. No special intraday return time is needed.
Ranking	• Leaderboard ranked by total portfolio value (same formula as above), refreshed every 60 s or via a refresh button. • Shown as "results provisional" until Day 7 finalisation, then frozen. • Equal values (to two decimal places) share the same competition rank. If a prize requires a single winner, a tie policy must be published before the game — recommend sharing the prize?

Supporting detail — MAT Stock Market — Technical Specification	
Data Preparation	• Uses real school data. Prepare a CSV/sheets data pack in advance using historical days of real data (not only Math Week), so no data collection is needed on event days. • 5 indices, with potential to add individual stocks. • Example — SOC recommendation index = 100 × (likes + comments + saves) / followers at measurement (posts must be published before Math Week).
Daily Price Update	Before the event, create a baseline B (average) for each index. Each day the hidden historical value O^d is revealed and: • RawChangePercent = 100 × (O^d − B) / B • DailyReturn = clamp(RawChangePercent × sensitivity, −8, +8) • TomorrowPrice = round(TodayPrice × (1 + DailyReturn / 100), 2) Sensitivity is locked before the event. The ±8% clamp keeps abnormal data from dominating the game. Because the formula uses real historical data, participants cannot know tomorrow's answer.
Worked Example	B = 1000 and the hidden next-day value is 1120. When revealed, RawChangePercent = 12%; with sensitivity 0.5, DailyReturn = +6%. A $100 LIB index becomes $106, so players holding it gain.
Trade Cost Formulas	• BUYCOST = quantity × price + max($1, round(0.01 × quantity × price, 2)) • SELLPROCEEDS = quantity × price − max($1, round(0.01 × quantity × price, 2))

***Supporting detail* — MAT Stock Market — Technical Specification**	
	• A buy is rejected if cash is insufficient; a sell is rejected if units held are insufficient.
Portfolio Value	Value = cash + Σ (units held × latest published price). The same formula is used for the live leaderboard and for the final ranking at market close.

MAT Mini Event

Dates, times and venues follow the approved proposal (Section 6C). Mini events are supervised, run at announced times and need more than three committee members each.

MAT Movie Night	
Overview	A movie screening featuring a film about mathematics, intellectual ability and personal development. The selected movie, *Good Will Hunting*, follows a young man with exceptional mathematical talent and explores themes of mathematics, education, relationships and personal growth. Held on the final day, immediately before the Closing Ceremony.
Film	*Good Will Hunting* (1997, 126 min)
Date & Time	Saturday 24 October 2026, 1.00 pm – 3.00 pm; followed by the Closing Ceremony (prize-awarding, closing remarks, event summary) 3.00 pm – 3.30 pm.
Venue	Mini Cinema
Person in Charge	Lee Shuan Fung

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty

MAT Expository Talk — Tiling and Aperiodic Tiling	
Overview	A talk delivered by the Mathematics Department presenting one accessible piece of mathematics to a general audience, followed by a hands-on workshop in the same room. The subject is **tiling**: how shapes fill the plane, why patterns repeat, and the shapes that never repeat. It carries the Math Week core message directly — the mathematics is visible in wallpaper, batik and songket motifs, Islamic star patterns and Escher prints, and it connects to the Exhibition (I.B.1 orangutan tiling, I.B.2 mathematics in wallpaper across cultures, and the cultural display cards).
Speaker	Dr. Darren Ong, Department of Mathematics. {speaker background for the introduction}
Topic	**Tiling and aperiodic tiling.** Suggested arc for the talk: 5. **What a tiling is** — tiles, edge-to-edge tilings, the three regular and eight semi-regular tilings. 6. **Why patterns repeat** — translation lattices, the 17 wallpaper groups, and the crystallographic restriction: a repeating pattern can only have 2-, 3-, 4- or 6-fold rotation, never 5-fold. 7. **Tiling in art and culture** — Islamic girih and star patterns, Escher, batik and songket motifs. 8. **Aperiodic sets** — Wang tiles and the question of whether a tile set forces non-repetition; Penrose's two tiles (kite and dart, or the two rhombs) with their matching rules; inflation/substitution, and the golden ratio appearing as the limiting ratio of the two tile types. 9. **Quasicrystals** — Shechtman's 1982 diffraction pattern with forbidden 5-fold symmetry (Nobel Prize 2011), and the cut-and-project view of Penrose tilings. 10. **The einstein problem, solved in 2023** — the **hat** monotile (Smith, Myers, Kaplan & Goodman-Strauss, March 2023) and the **spectre** (May 2023), a single tile that tiles the plane only aperiodically, without needing reflections. Reference: An aperiodic monotile. {final title}
Date & Time	Wednesday 21 October 2026, 7.00 pm – 9.00 pm
Venue	Library, 1st Floor
Programme	Talk first, workshop after, in the same room: 1. **7.00 – 7.15 pm — Arrival.** Seating; workshop tiles, scissors and laptops already laid out on the tables. 2. **7.15 – 7.20 pm — Opening.** Welcome and speaker introduction (MAT Student Council). 3. **7.20 – 8.00 pm — Talk (40 min).** Tiling and aperiodic tiling, as above. 4. **8.00 – 8.10 pm — Q&A (10 min).**

MAT Expository Talk — Tiling and Aperiodic Tiling	
	5. **8.10 – 8.50 pm — Workshop (40 min).** Participants move to the stations; each station is run by one committee member. See the **workshop suggestions** table below. 6. **8.50 – 9.00 pm — Showcase and close.** Finished tilings are laid out on one table (and the best go onto the Exhibition wall for the rest of the week), group photo, closing remarks.
Workshop	Run as parallel stations so participants choose what they want to try; a mix of physical (cut-and-fit) and digital (laptop or touchscreen) options is listed in the **workshop suggestions** table below. Decision: which stations to run (3–4 fit comfortably in 40 minutes), how many participants per station, and whether participants keep what they make or it joins the Exhibition wall.
Person in Charge	Jason William Teja

Supporting detail — MAT Expository Talk — Workshop Suggestions (tiling)	
A. Physical stations (cut, fit and take home)	
P1 · Penrose tiling	Tile a tray with Penrose kites and darts (or the two rhombs). The matching arcs must line up — that single rule is what forbids a repeating pattern. • **Shows:** matching rules, forced non-periodicity; count the tiles used — kites to darts approaches the golden ratio $\varphi \approx 1.618$. • **Needs:** ~100 tiles in card or acrylic, arcs printed on both faces.
P2 · The hat and the spectre	Cut out hat tiles from the printable sheet and build the largest patch the table allows; then try the spectre, whose wavy edges remove the need for reflected copies. • **Shows:** the 2023 solution of the einstein problem; roughly one hat in seven must be a mirror image. • **Needs:** printed sheets (hats.pdf), scissors.
P3 · Make your own Escher tile	Start from a square or hexagon, cut a shape out of one side and tape it onto the opposite side (translation) or the adjacent side (rotation). The result still tiles; trace it repeatedly, decorate it into a creature, and add it to a communal wall. • **Shows:** why cut-and-slide preserves tiling; the link to the exhibition's orangutan tiling (I.B.1). • **Needs:** card, scissors, tape, colour pens. Easiest station — good for passers-by.
P4 · Girih / Islamic star pattern	Build a ten-fold star pattern, either by compass-and-straightedge construction or by fitting the five girih tiles (decagon, pentagon, hexagon, bowtie, rhombus) cut from card. • **Shows:** 5- and 10-fold symmetry in historic ornament — the same symmetry a periodic tiling cannot have. • **Needs:** compass, ruler, protractor or pre-cut girih tiles.

Supporting detail — MAT Expository Talk — Workshop Suggestions (tiling)	
P5 · Substitution by hand	Take the L-shaped chair tile: four copies make a bigger copy of the same shape. Draw three generations on graph paper and watch a whole tiling grow from one rule. • **Shows:** inflation/substitution, the engine behind Penrose and hat tilings. • **Needs:** graph paper, pencils; pre-cut chair tiles help.
P6 · Wang tiles	Edge-coloured squares that may not be rotated: try to tile a 5 × 5 frame so that touching edges match. • **Shows:** the origin of the aperiodicity question — whether a tile set tiles the plane at all is undecidable in general. • **Needs:** one printed set of coloured-edge squares per table.
P7 · Laser diffraction demo (optional, with PHY)	Shine a laser pointer through a transparency printed with a Penrose pattern; the spots on the wall form a ten-fold star. • **Shows:** why quasicrystals were recognised from their diffraction patterns (Shechtman, Nobel 2011). • **Needs:** laser pointer, printed transparency, a dark corner. Decision: keep or drop — needs PHY help.
B. Digital stations (browser only, no installation)	
D1 · HatViz	Kaplan's interactive hat app: grow patches, step through the substitution levels, recolour and export a PNG of what you built. cs.uwaterloo.ca/~csk/hat/app.html • **Shows:** hierarchical structure — every patch sits inside a larger one.
D2 · Spectre app	The same for the chiral monotile: build a patch of the spectre and save it. cs.uwaterloo.ca/~csk/spectre/app.html • **Shows:** a single tile with no reflected copies at all.
D3 · Polypad hat table	Drag-and-drop hat tiles on a touchscreen — the gentlest digital option, and it runs on the Math Week touchscreen station. mathigon.org/polypad • **Shows:** the same puzzle as P2 without the scissors; good for a queue.
D4 · Tactile — design an isohedral tile	Pick one of the 93 isohedral tiling types, drag its edges into any shape you like and watch the whole tiling update; export SVG to print or cut. isohedral.ca/software/tactile • **Shows:** the classification of tilings by how a tile meets its neighbours; feeds P3 with a printable tile.
D5 · Wallpaper-group painting (this one is the custom design)	Draw a single stroke and see it repeated by the symmetry group you chose; switch groups and compare. eschersket.ch or KaleidoPaint. • **Shows:** the 17 wallpaper groups as something you can feel rather than memorise.
D6 · Code a substitution tiling	Fifteen lines of p5.js or Python turtle that apply a substitution rule (chair or Penrose) a few times and draw the result. editor.p5js.org — nothing to install. • **Shows:** recursion as the mechanism behind aperiodic order; aimed at CS/AI and Data Science students.

Supporting detail — **MAT Expository Talk — Workshop Suggestions (tiling)**	
D7 · Cut-and-project in GeoGebra	Draw a square lattice, lay a strip of irrational slope across it and project the points inside the strip onto a line: a one-dimensional quasicrystal with a Fibonacci pattern of gaps. geogebra.org • **Shows:** where Penrose tilings come from — a slice of a higher-dimensional periodic lattice.
D8 · Tilings Encyclopedia	Browse a catalogue of substitution tilings with their rules and pictures. tilings.math.uni-bielefeld.de • **Shows:** how many of these objects there are; useful as the station's reference screen.

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty

Library Data Talk	
Overview	A short talk presenting and discussing the Library data behind the MAT Stock Market — footfall (LIB), books borrowed and returned (BOOK), donated volumes (DONA) and social-media engagement (SCM) — and the insights found when the historical data pack was analysed. The analysed data is also shown on canvases in the Exhibition. Scheduled right after the Opening Ceremony and guided tour.
Presenter	Kow Chun Xiang
Topic	Analysis of the Library's operational data and how it drives the Stock Market indices. {final title}
Date & Time	Monday 19 October 2026, 2.30 pm – 3.00 pm
Venue	Library, 1st Floor
Person in Charge	Kow Chun Xiang

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty

Rubik Club Mini Booth	
Overview	A collaboration session with the Rubik Society, run as a mini booth next to the *Group Theory & The Rubik's Cube* exhibition so visitors can move from the mathematics on display to solving cubes hands-on. The booth activity itself (tiers, verification, points) is specified under Committee Based → Rubik Mini Booth.
Main Activity	Speed-solving demonstrations, beginner tutorials (first-layer method) and the timed-solve challenge linked to the Math Week leaderboard. {confirm with the Rubik Society}
Date & Time	Tuesday 20 October 2026, 7.00 pm – 9.00 pm
Venue	Library, 1st Floor
Person in Charge (contact)	Cher Ming Sian — liaison with the Rubik Society

Materials (if applicable)

Item	Qty/Activity	Remarks	Total Qty

Event Logistics Details

Master Material List

Available here:

Item	Remarks	Total Qty

Master Display List

Available here: 2610_MathWeek_DisplayList_PrepTracker.xlsx

Other Headers to include (if applicable):

- Theme
- Introduction (for talk, including speaker's background)
- Purpose of Event
- Target Group / Participant and Number of Attendees/ Participation Details
- Organizing Committee
- VVIP / VIP list
- Layout of Event / Floor Plan
- Logistics
- Financial Estimation
- Marketing and Publicity Method
- Project Timeline